

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE
Regular & Supplementary Winter Examination-2023

Course: B. Tech.

Semester : V

Branch : Electrical Engineering and allied

Subject Code & Name: Power Quality Issues (BTEEPLE504B)

Max Marks: 60

Date: 08/01/2024

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	Level/ CO	Marks
Q.1 Solve Any Two of the following.		12
A) Define power quality? Explain Power quality evaluation procedure	CO1	6
B) Explain short duration voltage variation with example	CO1	6
C) Define Total Harmonic Distortion. Explain the procedure for calculation the Total Harmonic Distortion(THD) due to disturbance in the power system	CO2	6
Q.2 Solve Any Two of the following.		12
A) Explain sources of transient overvoltage. Explain any two devices for overvoltage protection.	CO2	6
B) Explain sources of voltage sag also draw its characteristics when short circuit occurs on parallel feeder.	CO1	6
C) Describe Ferro-resonance. Draw graphical solution to linear LC circuit and nonlinear LC circuit.	CO3	6
Q.3 Solve Any Two of the following.		12
A) Explain the term harmonic phase sequences and Triplen harmonics.	CO4	6
B) Explain concept of point of common coupling	CO2	6
C) Explain any three devices used for controlling harmonics distortion.	CO1	6
Q.4 Solve Any Two of the following.		12
A) Explain principle of regulating the voltage also states common option for improving power system voltage regulation	CO1	6
B) Explain any four devices to maintain voltage regulation	CO2	6
C) Define flicker and explain sources of flicker.	CO3	6

Q. 5 Solve Any Two of the following.		12
A) Explain various power quality monitoring standards.	CO1	6
B) Describe online power quality data assessment also explain application of intelligent systems.	CO4	6
C) List various types of power quality measuring equipment and explain any three type power quality measuring system.	CO1	6
